$$= \sqrt{13^2 - 5^2}$$

$$= \sqrt{169 - 25} \quad = \quad \sqrt{144} = 12$$

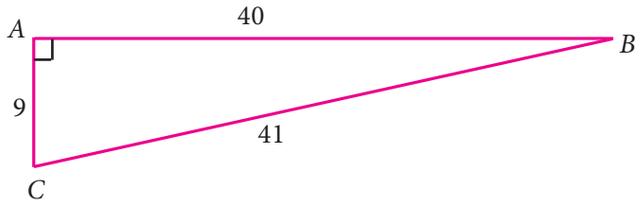
Therefore, $\sin\theta = \frac{AC}{BC} = \frac{12}{13}$; $\cos\theta = \frac{AB}{BC} = \frac{5}{13}$

$$LHS = \frac{2\sin\theta - 3\cos\theta}{4\sin\theta - 9\cos\theta} = \frac{2\times\frac{12}{13} - 3\times\frac{5}{13}}{4\times\frac{12}{13} - 9\times\frac{5}{13}} = \frac{\frac{24-15}{13}}{\frac{48-45}{13}} = \frac{9}{3} = 3 = RHS$$

Note: We can also take the angle 'θ' at the vertex 'C' and proceed in the same way.

Exercise 6.1

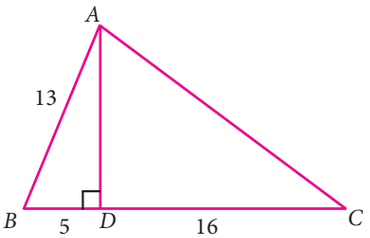
1. From the given figure, find all the trigonometric ratios of angle B.



2. From the given figure, find the values of

(i) $\sin B$ (ii) $\sec B$ (iii) $\cot B$

(iv) $\cos C$ (v) $\tan C$ (vi) $\operatorname{cosec} C$



3. If $2\cos\theta = \sqrt{3}$, then find all the trigonometric ratios of angle θ.

4. If $\cos A = \frac{3}{5}$, then find the value of $\frac{\sin A - \cos A}{2\tan A}$.

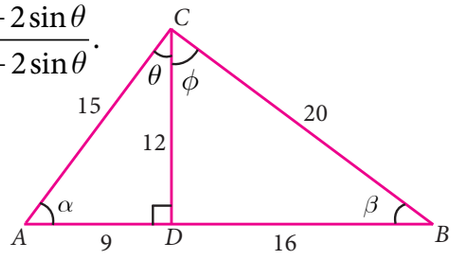
5. If $\cos A = \frac{2x}{1+x^2}$, then find the values of sin$A$ and tan$A$ in terms of x.

6. If $\sin\theta = \frac{a}{\sqrt{a^2+b^2}}$, then show that $b\sin\theta = a\cos\theta$.

7. If $3\cot A = 2$, then find the value of $\frac{4\sin A - 3\cos A}{2\sin A + 3\cos A}$.

8. If $\cos\theta : \sin\theta = 1:2$, then find the value of $\frac{8\cos\theta - 2\sin\theta}{4\cos\theta + 2\sin\theta}$.

9. From the given figure, prove that $\theta + \phi = 90°$. Also prove that there are two other right angled triangles. Find $\sin\alpha$, $\cos\beta$ and $\tan\phi$.



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